

Computational Reproducibility and Data Usage Guide for Approach-Region Dynamics and Rate Turnover in Thermally Activated Barrier Crossing

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Overview

This document describes the datasets, computational tools, and analysis procedures required to reproduce the calculations presented in the manuscript:

Approach-Region Dynamics and Rate Turnover in Thermally Activated Barrier Crossing.

The repository contains scripts and datasets required to reproduce:

- approach-region damping ratio χ_0 calculations via three estimator routes
- dynamical regime classifications (underdamped / near-critical / overdamped)
- cross-domain statistical clustering analysis (Test 8)
- catalytic dataset construction used in the manuscript figures

All tools are implemented in Python and operate directly on the included CSV datasets. No proprietary software is required. The canonical Python module is `symc_methods.py`, available at: github.com/SymCUniverse/Chemistry.

Epistemic note on the central quantity

The manuscript was revised to correct errors identified during peer review of previous research. The central quantity is the *approach-region* damping ratio:

$$\chi_0 = \frac{\Gamma_{\text{eff}}}{2\Omega_0} \quad (1)$$

where Ω_0 is the precursor-well approach frequency ($\Omega_0^2 > 0$, stable coordinate) and Γ_{eff} is the total projected dissipation. The exceptional-point boundary at $\chi_0 = 1$ belongs to the stable precursor-well generator, not the inverted barrier-top coordinate. Any earlier version of this guide or code that defines the central quantity as $\chi = \Gamma_{\text{eff}}/(2\Omega_{\text{TS}})$ using the barrier-top curvature frequency Ω_{TS} reflects the pre-revision framework and should be superseded by this document.

Route B was also corrected: the damping rate scales as $\Gamma_{\text{eff}} = \Omega_s^2 \tau_L$ (Debye-Drude friction, proportional to τ_L), not as $\Gamma_{\text{eff}} = 1/\tau_L$. The PCET isotope shift is $\chi_0^D = \sqrt{2} \chi_0^H$ as an upper bound for proton-dominated coordinates (solvent-dressed coordinates show a smaller shift, but the direction is the same: H→D raises χ_0).

1 Core χ_0 Computation Methods

All χ_0 calculations are implemented in `sync_methods.py`. This module provides three estimator routes connecting χ_0 to spectroscopic, solvent-dynamics, or electronic-friction observables.

1.1 Route A: Spectroscopic Estimator

Function: `chi_route_a(delta_nu_L_cm1, nu_cm1)`

$$\chi_0^{\text{spec}} = \frac{\Delta\tilde{\nu}_L}{2\tilde{\nu}} \quad (2)$$

where $\Delta\tilde{\nu}_L$ is the *Lorentzian* component of the IR or Raman linewidth only (not the raw FWHM). Valid when the Lorentzian fraction exceeds 50%; invalid when Gaussian (inhomogeneous) broadening dominates.

1.2 Route B: Solvent-Relaxation Estimator

Important: Absolute Route B values depend quadratically on the poorly constrained solvent librational frequency Ω_s and can span a 10-fold range. Absolute values should not be used for regime classification. The operationally reliable quantity is the *relative* damping ratio.

Relative estimator (preferred)

Function: `chi_route_b_relative(tau_L_a_ps, tau_L_b_ps)`

$$\frac{\chi_{\text{ET}}^{(A)}}{\chi_{\text{ET}}^{(B)}} \approx \frac{\tau_L^{(A)}}{\tau_L^{(B)}} \quad (3)$$

Valid when m_{eff} , λ , and ΔQ are constant (fixed donor–acceptor pair) and the compared solvents have similar librational frequencies ($\Omega_s^{(A)} \approx \Omega_s^{(B)}$, i.e. same-class solvents such as nitriles or carbonates). For cross-class comparisons, multiply by $(\Omega_{s,A}/\Omega_{s,B})^2$. Larger τ_L gives larger χ_0 (more overdamped).

Absolute estimator (for screening only)

Function: `chi_route_b_absolute(tau_L_ps, lambda_ev, delta_q_ang, m_eff_amu, omega_s_cm)`

Physical basis: $\Gamma_{\text{eff}} = \Omega_s^2 \tau_L$ (van der Zwan & Hynes 1982).

$$\chi_0 = \frac{\Omega_s^2 \tau_L}{2\Omega_0}, \quad \Omega_0 = \sqrt{\frac{2\lambda}{m_{\text{eff}} (\Delta Q)^2}} \quad (4)$$

Always evaluate over a range of $m_{\text{eff}} \in \{50, 100, 200, 500\}$ amu and $\Omega_s \in \{50, 100, 150\}$ cm^{−1} and report the full bracket. A regime assignment is reliable only if the entire bracket falls within a single regime.

1.3 Route C: Electronic Friction Estimator

Function: `chi_route_c(gamma_cat_ev, nu_cm1, gamma_int_ev)`

$$\chi_{\text{DFT}} = \frac{\Gamma_{\text{cat}} + \Gamma_{\text{int}}}{2hc\tilde{\nu}}, \quad hc = 1.23984 \times 10^{-4} \text{ eV} \cdot \text{cm} \quad (5)$$

Γ_{cat} is the **Newns–Anderson hybridization energy broadening** $2\pi|V_k|^2\rho(\varepsilon_F)$ in eV — an electronic energy width, not a vibrational friction coefficient. The formula is the ratio of two energies and is strictly dimensionless. Route C values are approach-region (ground-state geometry) estimates; the saddle-region χ_{TS} is larger for systems where the reactive bond softens toward the transition state.

1.4 PCET Isotope Function

Function: `chi_pcet_isotope(chi_H)`

$$\chi_0^D = \sqrt{2} \chi_0^H \approx 1.41 \chi_0^H \quad (\text{upper bound; proton-dominated limit}) \quad (6)$$

H→D substitution doubles the proton mass, *lowers* Ω_0 by $\sqrt{2}$, and therefore *raises* χ_0 by $\sqrt{2}$ in the limit where the proton mass dominates the effective coordinate mass ($m_{\text{eff}} \approx m_{\text{proton}}$). In PCET systems where solvent reorganization contributes significantly to the effective mass, the actual frequency shift is smaller and $\chi_0^D < \sqrt{2} \chi_0^H$. The factor $\sqrt{2}$ is an upper bound; the qualitative direction (D raises χ_0) is model-independent.

2 Approach-Region χ_0 Evaluation

Input dataset: `ts_chi_input.csv`

Processing script: `ts_chi_tools.py`

Output dataset: `chi_ts_results.csv`

The script reads the dataset and calculates χ_0 values via the appropriate estimator route, then classifies each system into dynamical regimes. The term “ts” in the filename refers to systems near a transition state; the χ_0 values computed are approach-region (precursor-well) estimates, not saddle-point values.

χ_0 Range	Dynamical Regime
$\chi_0 < 0.8$	Underdamped (oscillatory; recrossing-prone)
$0.8 \leq \chi_0 \leq 1.3$	Near-critical (fastest monotonic commitment)
$\chi_0 > 1.3$	Overdamped (monotonic but sluggish)

3 Cross-Domain Statistical Test (Test 8)

Script: `Test8_pValueCal.py`

The statistical test evaluates whether optimal systems across independent chemical domains cluster near $\chi_0 \approx 1$ by chance under a log-uniform prior. No external dataset is required; all inputs are hardcoded in the script.

$$0.8 \leq \chi_0 \leq 1.3 \quad (7)$$

assuming a log-uniform distribution across

$$10^{-4} \leq \chi_0 \leq 10^1. \quad (8)$$

For $N = 3$ candidate anchor systems the calculation produces:

- Single-system probability $\approx 4.22\%$
- Combined probability $\approx 7.50 \times 10^{-5}$

Prospective status. Test 8 is currently a prospective statistical sensitivity demonstration, not a formal validation. Only the catalysis domain (K-promoted Fe(111), $\chi_{\text{DFT}} \approx 1.28$ via Route C) provides a direct numerical χ_0 estimate; this value falls within the near-critical operational window $[0.8, 1.3]$. The electron-transfer and PCET anchors rest on structural (PGH turnover argument) and mechanistic (near-zero activation volume) evidence respectively, not on computed χ_0 values. Test 8 becomes a formal statistical test when at least one direct numerical χ_0 estimate is available in each of the three domains. The prior robustness sweep confirms $p < 0.05$ across all reasonable prior bounds ($[10^{-5}, 10^1]$ to $[10^{-4}, 10^3]$) and window widths ($[0.75, 1.35]$ to $[0.85, 1.25]$).

4 Catalysis Dataset Construction

Script: `oc20_n2.catalog.py`

Output dataset: `oc20_n2.catalog.csv`

This script extracts N₂ adsorption systems from the Open Catalyst Project (OC20) metadata and compiles the catalytic screening dataset. Route C χ_0 values in the dataset are ground-state approach-region estimates; the OC20 cohort χ values are deferred to future work.

5 Periodic Table χ_0 Proxy Dataset

Dataset: `periodic_table_chi_proxy.csv`

This dataset contains electron-phonon coupling λ_{ep} and Debye temperature Θ_D values used to compute the lattice damping proxy

$$\chi_{\text{lattice}} = \frac{\pi \lambda_{ep} T}{2 \Theta_D}. \quad (9)$$

χ_{lattice} is **not** χ_0 ; it is a substrate-damping proxy. Values are derived from McMillan (1968) and Allen-Dynes (1975) tabulations and are not imposed by parameter selection. The dataset covers $Z = 1-94$ with literature values and $Z > 94$ with block-average proxies as noted in Figure 3.

6 Figure Generation Scripts

All figure scripts require `matplotlib` with the `Agg` backend (`pip install matplotlib`) and `numpy/pandas`. No display or GUI is needed. Run from the repository root; outputs write to the current directory.

Figure 1 — Dynamical Efficiency Map

Script: `generate_fig1_cce_v9.py`

Outputs: `Fig1DynamicalEfficiencyTracker.png`, `.svg`, `_data.csv`

Generates the reduced symmetric turnover schematic $\eta(\chi) = 2\chi/(1 + \chi^2)$ with Route C Fe/N₂ data points on the main panel and the TEMPO Route B relative series in the inset. All Route C χ values are computed internally via $\chi_{\text{DFT}} = \Gamma_{\text{cat}}/(2hc\tilde{\nu})$ using the hardcoded Fe surface parameters; no external dataset is required.

```
python generate_fig1_cce_v9.py
```

Figure 2 — BEP Volcano Outliers (Test 5)

Script: `generate_fig2_corrected.py`

Outputs: `Fig2_FeBEPTest5.png`, `.svg`

Generates the BEP deviation schematic with the four Fe surface data points. K-promoted Fe(111) is placed at the Route C computed value $\chi_{\text{DFT}} = 1.2825$. The over-promoted Fe point ($\chi = 2.5$) is schematic; no Route C value is available for that loading. The predicted dynamical enhancement curve is a Gaussian in $\log \chi$ space centred at $\chi = 1$, included for visual illustration only.

```
python generate_fig2_corrected.py
```

SVG note. All text in the SVG is rendered as editable text nodes (`svg.fonttype = 'none'`). Mathematical symbols in annotations and axis labels are plain ASCII or Unicode. If a symbol renders incorrectly in Illustrator, set the affected text node to Cambria Math, which covers the full mathematical Unicode block reliably.

Figure 3 — Standard-State Lattice χ Map

Script: `generate_fig3_corrected.py`

Outputs: `Fig3_Elemental_Map.png`, `Fig3_Elemental_Map_editable.svg`

Computes $\chi_{\text{lattice}} = \pi \lambda_{ep} T / (2\Theta_D)$ for all 118 elements using hardcoded McMillan/Allen–Dynes values ($Z=1-94$) and block-average proxies ($Z>94$). The editable SVG assigns a unique `id` attribute to each element label group (`label_SYM_ZN`, e.g. `label_Fe_Z26`) and to the $\chi = 1$ line annotation (`chi_eq_1_line`), enabling label repositioning in Illustrator without disturbing the data layer.

```
python generate_fig3_corrected.py
```

7 Minimal Reproduction Workflow

Compute approach-region χ_0 values:

```
python ts_chi_tools.py ts_chi_input.csv
```

Run the cross-domain clustering test (prospective):

```
python Test8_pValueCal.py
```

Regenerate all three figures:

```
python generate_fig1_cce_v9.py
python generate_fig2_corrected.py
python generate_fig3_corrected.py
```

Optionally regenerate the OC20 catalytic screening dataset (requires internet access; χ values are deferred to future work):

```
python oc20_n2_catalog.py
```

Data Availability

All datasets, scripts, and templates required to reproduce the calculations presented in the manuscript are included in the repository. No proprietary data or external software dependencies are required. Code (`symc_methods.py`) and all input parameters are available at github.com/SymCUniverse/Chemistry. Contact: NateChristensen@SymCUniverse.com.